

COSC 594/650

Primal–Dual Method and Convergence

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Dual method and convergence

$$x_k \in \operatorname{argmin}_{x \in X} L(x, \lambda_k), \quad \lambda_{k+1} = [\lambda_k + \alpha(Ax_k - b)]_+$$

- EXACT INNER SOLVE GIVES A DUAL SUPERGRADIENT
- STRONG CONVEXITY MAKES THE DUAL SMOOTH
- PRIMAL ITERATES MAY BE INFEASIBLE

Assumption	Step size	Dual gap	Primal error
σ -strongly convex f_0	$\alpha \leq \sigma / \ A\ _2^2$	$O(1/K)$	$O(1/\sqrt{K})$
$Ax = b, \quad AA^\top \succ 0$	$\alpha = 1/\lambda_{\max}$	$(1 - 1/\kappa)^{2k}$	$(1 - 1/\kappa)^k$
convex only	$\alpha = R/(G\sqrt{K})$	$O(K^{-1/2})$	no general bound

QUESTION: Can we avoid the inner minimization?

Strong duality and saddle points

Primal problem: $\min_{x \in X} f_0(x)$ subject to $Ax \leq b$

Lagrangian: $L(x, \lambda) = f_0(x) + \lambda^\top (Ax - b)$

■ STRONG DUALITY WITH ATTAINED OPTIMA

$$f_0(x^*) = p^* = d^* = g(\lambda^*)$$

★ x^* is primal feasible and $\lambda^* \geq 0$

■ EQUIVALENT SADDLE-POINT CONDITION

$$L(x^*, \lambda) \leq L(x^*, \lambda^*) = p^* \leq L(x, \lambda^*), \quad x \in X, \lambda \geq 0$$

★ x^* minimizes $L(\cdot, \lambda^*)$; λ^* maximizes $L(x^*, \cdot)$

Strong duality + Primal and dual attainment $\iff$ A saddle point exists

Dual method versus primal–dual method

Saddle-point problem: $\underset{x \in X}{\text{minimize}} \quad \underset{\lambda \geq 0}{\text{maximize}} \quad L(x, \lambda)$

Lagrangian: $L(x, \lambda) = f_0(x) + \lambda^\top (Ax - b)$

■ DUAL METHOD

$$x_k \in \underset{x \in X}{\operatorname{argmin}} L(x, \lambda_k)$$

$$\lambda_{k+1} = [\lambda_k + \alpha(Ax_k - b)]_+$$

- ★ exact primal minimization
- ★ $Ax_k - b$ – dual supergradient

■ PRIMAL–DUAL METHOD

$$x_{k+1} = \Pi_X(x_k - \alpha \nabla_x L(x_k, \lambda_k))$$

$$\lambda_{k+1} = [\lambda_k + \alpha(Ax_k - b)]_+$$

- ★ one primal gradient step
- ★ both updates use (x_k, λ_k)

Primal update runs one gradient step

Projected primal–dual gradient method

Setting: L differentiable and convex–concave; X, Λ closed convex

Initialization: $(x_0, \lambda_0) \in X \times \Lambda$, and stepsize $\alpha > 0$

$$x_{k+1} = \Pi_X(x_k - \alpha \nabla_x L(x_k, \lambda_k))$$

$$\lambda_{k+1} = \Pi_\Lambda(\lambda_k + \alpha \nabla_\lambda L(x_k, \lambda_k))$$

■ LAGRANGIAN GRADIENTS

$$\nabla_x L(x, \lambda) = \nabla f_0(x) + A^\top \lambda$$

$$\nabla_\lambda L(x, \lambda) = Ax - b$$

- ★ simultaneous descent and ascent
- ★ one evaluation of each partial gradient, plus two projections

Without an exact inner solve: $Ax_k - b$ needs not be a supergradient of g

Example: one inequality constraint

$$\begin{array}{ll} \underset{x}{\text{minimize}} & \frac{1}{2} x^2 \\ \text{subject to} & 1 - x \leq 0 \end{array} \quad \begin{array}{l} L(x, \lambda) = \frac{1}{2} x^2 + \lambda(1 - x) \\ x^* = \lambda^* = 1, \quad X = \mathbb{R}, \quad \Lambda = \mathbb{R}_+ \end{array}$$

$$x_{k+1} = x_k - \frac{1}{2} (x_k - \lambda_k)$$

$$\lambda_{k+1} = \left[\lambda_k + \frac{1}{2} (1 - x_k) \right]_+, \quad (x_0, \lambda_0) = (0, 0)$$

k	x_k	λ_k	$[1 - x_k]_+$	$x(\lambda_k) = \lambda_k$
0	0	0	1	0
1	0	1/2	1	1/2
2	1/4	1	3/4	1
3	5/8	11/8	3/8	11/8
4	1	25/16	0	25/16

x_k lags behind $x(\lambda_k)$, and λ_k overshoots λ^*

Why the last iterate can fail

Bilinear saddle: $\underset{x \in \mathbb{R}}{\text{minimize}} \underset{\lambda \in \mathbb{R}}{\text{maximize}} \ x\lambda$

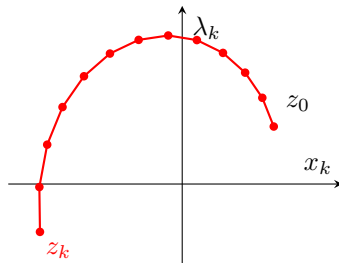
■ SIMULTANEOUS UPDATES

$$x_{k+1} = x_k - \alpha \lambda_k$$

$$\lambda_{k+1} = \lambda_k + \alpha x_k$$

■ THE CROSS TERMS CANCEL

$$\begin{aligned} x_{k+1}^2 + \lambda_{k+1}^2 &= (x_k - \alpha \lambda_k)^2 + (\lambda_k + \alpha x_k)^2 \\ &= (1 + \alpha^2)(x_k^2 + \lambda_k^2) \end{aligned}$$



$$x_k^2 + \lambda_k^2 = (1 + \alpha^2)^k (x_0^2 + \lambda_0^2) \longrightarrow \infty \quad \text{if } (x_0, \lambda_0) \neq (0, 0), \alpha > 0$$

The primal–dual gap and assumptions

■ BOUNDED LAGRANGIAN GRADIENTS

$$\|\nabla_x L(x, \lambda)\|^2 + \|\nabla_\lambda L(x, \lambda)\|^2 \leq G^2$$

- ★ X, Λ nonempty compact convex
- ★ L differentiable and convex–concave

■ UNIFORM INITIAL-DISTANCE BOUND

$$D_k(x, \lambda) = \|x_k - x\|^2 + \|\lambda_k - \lambda\|^2$$

- ★ $R^2 = \max_{(x, \lambda) \in X \times \Lambda} D_0(x, \lambda)$ – distance to every comparator

■ PRIMAL–DUAL GAP

$$\text{Gap}(x, \lambda) = \max_{\lambda' \in \Lambda} L(x, \lambda') - \min_{x' \in X} L(x', \lambda)$$

$$\text{Gap}(x, \lambda) = 0 \iff (x, \lambda) \text{ is a saddle point on } X \times \Lambda$$

One-step analysis: the two projection bounds

Comparator: any $(x, \lambda) \in X \times \Lambda$

■ PRIMAL DESCENT

$$\begin{aligned}\|x_{k+1} - x\|^2 &\leq \|x_k - \alpha \nabla_x L(x_k, \lambda_k) - x\|^2 \\ &= \|x_k - x\|^2 - 2\alpha \langle \nabla_x L(x_k, \lambda_k), x_k - x \rangle + \alpha^2 \|\nabla_x L(x_k, \lambda_k)\|^2\end{aligned}$$

■ DUAL ASCENT

$$\begin{aligned}\|\lambda_{k+1} - \lambda\|^2 &\leq \|\lambda_k + \alpha \nabla_\lambda L(x_k, \lambda_k) - \lambda\|^2 \\ &= \|\lambda_k - \lambda\|^2 + 2\alpha \langle \nabla_\lambda L(x_k, \lambda_k), \lambda_k - \lambda \rangle + \alpha^2 \|\nabla_\lambda L(x_k, \lambda_k)\|^2\end{aligned}$$

One-step inequality for primal–dual updates

Comparator: any $(x, \lambda) \in X \times \Lambda$

■ CONVEXITY IN x , CONCAVITY IN λ

$$L(x_k, \lambda_k) - L(x, \lambda_k) \leq \langle \nabla_x L(x_k, \lambda_k), x_k - x \rangle,$$

$$L(x_k, \lambda) - L(x_k, \lambda_k) \leq \langle \nabla_\lambda L(x_k, \lambda_k), \lambda - \lambda_k \rangle.$$

■ ADD THE PROJECTION BOUNDS

$$D_{k+1}(x, \lambda) \leq D_k(x, \lambda) - 2\alpha [L(x_k, \lambda) - L(x, \lambda_k)] + \alpha^2 G^2$$

★ no sufficient-ascent guarantee – the iterates move simultaneously

$$\implies L(x_k, \lambda) - L(x, \lambda_k) \leq \frac{D_k(x, \lambda) - D_{k+1}(x, \lambda)}{2\alpha} + \frac{\alpha G^2}{2}$$

Convergence analysis for the primal–dual method

Averaged iterates: $\bar{x}_K = K^{-1} \sum_{k=0}^{K-1} x_k, \quad \bar{\lambda}_K = K^{-1} \sum_{k=0}^{K-1} \lambda_k$

Comparator bound: $D_0(x, \lambda) \leq R^2$

■ TELESCOPE

$$\frac{1}{K} \sum_{k=0}^{K-1} [L(x_k, \lambda) - L(x, \lambda_k)] \leq \frac{R^2}{2\alpha K} + \frac{\alpha G^2}{2}$$

■ AVERAGE AND TAKE THE EXTREMA

$$L(\bar{x}_K, \lambda) - L(x, \bar{\lambda}_K) \stackrel{\text{Jensen}}{\leq} \frac{1}{K} \sum_{k=0}^{K-1} [L(x_k, \lambda) - L(x, \lambda_k)]$$

★ α is set for a planned horizon K – a fixed α leaves an $\alpha G^2/2$ floor

$$\Rightarrow \text{Gap}(\bar{x}_K, \bar{\lambda}_K) \leq \frac{R^2}{2\alpha K} + \frac{\alpha G^2}{2} \stackrel{\alpha=R/(G\sqrt{K})}{=} \frac{RG}{\sqrt{K}}$$

Primal feasibility and objective accuracy

Dual ball: $\Lambda = \{\lambda \geq 0 : \|\lambda\| \leq B\}$, where $B \geq 2\|\lambda^*\|$

Accuracy: $\text{Gap}(\bar{x}_K, \bar{\lambda}_K) \leq \epsilon$

■ THE GAP ALREADY CONTAINS THE VIOLATION

$$\max_{\lambda \in \Lambda} L(\bar{x}_K, \lambda) = f_0(\bar{x}_K) + B\|[A\bar{x}_K - b]_+\|, \quad \min_{x \in X} L(x, \bar{\lambda}_K) \leq p^*$$

$$\star f_0(\bar{x}_K) - p^* + B\|[A\bar{x}_K - b]_+\| \leq \epsilon$$

■ THE SADDLE POINT BOUNDS f_0 FROM BELOW

$$f_0(\bar{x}_K) - p^* \geq -\|\lambda^*\| \|[A\bar{x}_K - b]_+\|$$

$$\text{Objective gap: } |f_0(\bar{x}_K) - p^*| \leq \epsilon$$

$$\text{Constraint violation: } \|[A\bar{x}_K - b]_+\| \leq \frac{2\epsilon}{B}$$

Choosing the dual radius from a Slater point

Slater point: $\tilde{x} \in X$ with $A\tilde{x} - b \leq -\delta \mathbf{1}$, $\delta > 0$

Lower bound: any computable $\ell \leq p^\star$

■ THE SLATER POINT BOUNDS THE MULTIPLIER

$$p^\star = g(\lambda^\star) \leq L(\tilde{x}, \lambda^\star) \leq f_0(\tilde{x}) - \delta \|\lambda^\star\|_1$$

$$\star \quad \|\lambda^\star\| \leq \|\lambda^\star\|_1 \leq \frac{f_0(\tilde{x}) - \ell}{\delta}$$

■ PROJECTION ONTO THE DUAL BALL

$$\Pi_\Lambda(v) = \frac{[v]_+}{\max\{1, \|[v]_+\|/B\}}$$

$$B = \frac{2(f_0(\tilde{x}) - \ell)}{\delta} \implies \|[A\bar{x}_K - b]_+\| \leq \frac{2\epsilon}{B}$$

Primal–dual method and convergence

$$x_{k+1} = \Pi_X(x_k - \alpha \nabla_x L(x_k, \lambda_k)), \quad \lambda_{k+1} = \Pi_\Lambda(\lambda_k + \alpha \nabla_\lambda L(x_k, \lambda_k))$$

- ONE PRIMAL STEP REMOVES THE EXACT INNER SOLVE
- THE GAP GUARANTEE BELONGS TO THE AVERAGED PAIR
- A LARGE ENOUGH DUAL BALL CONTROLS PRIMAL VIOLATION

Method	Work per iteration	Gap bound	Rate
Smooth dual	exact solve	last dual	$O(1/K)$
Nonsmooth dual	exact solve	best dual	$O(1/\sqrt{K})$
Primal–dual	gradient step	averaged saddle	$O(1/\sqrt{K})$

Questions?

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